Dhimitrios Duka

Incoming Ph.D. Student, CVML Group, University of Tübingen
M.Sc. Computer Science, Saarland University

EDUCATION

University of Tübingen Starting 12/2026
Incoming Ph.D. Student, CVML Group, Tübingen AI Center
Advisor: Prof. Dr. Hilde Kuehne

Saarland University 10/2023 – 10/2026
M.Sc. in Computer Science, GPA: 1.5 (1.0 best, 4.0 worst)
Thesis: Right on Time: Temporal Boundary Refinement for Improved Egocentric Visual-Language Pre-Training
Advisors: Dr. Anna Kukleva & Prof. Dr. Bernt Schiele

Polytechnic University of Tirana 10/2018 – 03/2022
B.Sc. in Computer Engineering, GPA: 9.7 (10.0 best, 5.0 worst; German equivalent: 1.1)
Thesis: A Neural Image Caption Generator for the Albanian Language
Advisors: Dr. Evis Trandafili & Prof. Dr. Elinda Kajo Meçe

PUBLICATIONS

[1] Siarhei Sheludzko, **Dhimitrios Duka**, Bernt Schiele, Hilde Kuehne, Anna Kukleva. 2026. MM-TS: Multi-Modal Temperature and Margin Schedules for Contrastive Learning with Long-Tail Data. In *Proceedings of the IEEE/CVF Winter Conference on Applications of Computer Vision* (WACV 2026). arXiv:2603.08202. [Oral Presentation]

[2] **Dhimitrios Duka**, Bernt Schiele, Hilde Kuehne, Anna Kukleva. 2026. TeMo: Temperature Modulation for Multimodal Contrastive Learning. *arXiv preprint* arXiv:2609.07540.

TALKS & PRESENTATIONS

- Invited Talk, **MM-TS**, Department of Computer Engineering, Middle East Technical University 04/2026
- Oral Presentation, **MM-TS**, WACV 2026, Tucson, Arizona 03/2026
- Seminar Talk, **Generative Models in Computer Vision**, Prof. Dr. Jan Eric Lenssen 08/2025
- Seminar Talk, **Explainable Machine Learning**, Dr. Jonas Fischer 08/2024

RESEARCH EXPERIENCE

Max Planck Institute for Informatics Saarbrücken, Germany
Research Assistant 11/2024 – 10/2026
Advisors: Dr. Anna Kukleva and Prof. Dr. Bernt Schiele

- **Master's Thesis:** Developed a timestamp-conditioned temporal boundary refinement framework for Ego4D, adapting vision-language models through supervised fine-tuning and difficulty-targeted GRPO; improved localization from **49.99 to 61.80 mIoU**.
- Built a benchmark of **12,814 manually annotated action intervals**, coordinating the annotation work of two student assistants.
- Investigated label-free GRPO objectives, recovering approximately 62% of reference-based GRPO gains without boundary annotations during the reinforcement-learning stage.
- Applied temporal refinement to the full EgoClip corpus, demonstrating that **pre-training on just 50% of the refined data outperformed the full original corpus** on downstream tasks.
- Developed temperature scaling methods for multimodal contrastive learning: **TeMo** (first author) and **MM-TS** (co-author, WACV 2026 Oral).

INDUSTRY EXPERIENCE

Sciant Tirana, Albania
Full-Stack Software Engineer 08/2023 – 10/2023

- Contributed to the development of a backend integration service for Protel, a hotel property management system, using Spring Boot, Node.js, and AWS (S3, DynamoDB) to enable reliable and scalable data synchronization.
- Implemented automated tests and technical documentation to improve system reliability, maintainability, and operational support.

DigitSapiens Tirana, Albania
Full-Stack Software Engineer 08/2021 – 07/2023

- Developed secure, end-to-end digital workflows for the Unified Patent Court's e-Justice platform, an online system for managing European patent litigation, adhering to strict reliability, privacy, and compliance requirements.
- Contributed to migrating a monolithic architecture to PostgreSQL-backed microservices, improving scalability and service integration.

AWARDS

- **SaarlandStipendium for International Students**, StudienStiftungSaar 2025, 2026
- **1st Place, Computer Graphics Rendering Competition**, Saarland University 02/2024

SELECTED PROJECTS

sCode – SLURM Cluster Manager [GitHub] 2025 – Present

- Created and actively maintaining an open-source VS Code extension with approximately 2,400 downloads, integrating SLURM job submission, monitoring, and cluster management directly into the editor.
- Implemented resource-aware SLURM script assistance, GPU and partition utilization monitoring, and detailed job diagnostics to simplify HPC workflows.

Multilingual LLM Evaluation & Fine-Tuning [GitHub] 03/2024

- Investigated low-resource language adaptation of XGLM-564M and GPT-2, evaluating language modeling performance and analyzing multilingual representations using PCA and t-SNE.
- Implemented and compared full fine-tuning with PEFT methods (BitFit, LoRA, IA3), achieving a 30.4% reduction in language modeling loss with full fine-tuning while exploring alternatives updating fewer than 0.28% of model parameters.

Brains at Play – EEG-Based Application Control [Private Repository] 03/2024

- Co-designed and developed a general-purpose brain-computer interface (BCI) enabling hands-free control of existing games and desktop applications through EEG-based SSVEP detection.
- Built a modular Python/PyQt5 framework integrating real-time EEG processing, FFT and FBCCA classification, customizable visual overlays, and programmable input actions.

Neural Image Captioning for Albanian [GitHub] 09/2021 – 03/2022

- Developed an end-to-end image captioning system for low-resource Albanian as part of my bachelor's thesis, constructing a vision-language dataset using pretrained neural machine translation.
- Trained a transfer-learning-based encoder-decoder model and optimized caption generation through decoding strategies, achieving a BLEU-4 score of 12.6.

TECHNICAL SKILLS

Machine Learning & AI: Large Language Models (LLMs), Vision-Language Models (VLMs), Multimodal Representation Learning, Contrastive Learning, Self-Supervised Learning, Generative Models, Supervised Fine-Tuning (SFT), Reinforcement Learning (GRPO), PEFT (LoRA, BitFit, IA3), Video-Language Understanding

Frameworks & Infrastructure: PyTorch, Hugging Face Transformers, vLLM, Weights & Biases, SLURM, Distributed & Multi-GPU Training, Docker, Kubernetes, AWS

Software Engineering: Python, TypeScript, Java, Node.js, React, PostgreSQL, CI/CD, Git

LANGUAGES

Albanian	Native	**English**	Full Professional Proficiency
Greek	Fluent Speaking; Limited Reading/Writing	**German**	A1